Can open citation data detect citation stacking? An audit against Clarivate's journal suppressions

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Abstract

Clarivate suppresses journals from the Journal Citation Reports when self-citations or concentrated citations from another journal ("citation stacking") distort their Journal Impact Factor. Its data are closed, so independent checks need open citation data. We audit how far fully open data can reproduce and anticipate these decisions. We built yearly journal citation networks from the OpenCitations snapshot of January 2024 (1.96 billion citations; 85,437 journals) and compared them with Clarivate's suppression lists for data years 2015–2023, collected from primary sources. Where a journal's reference lists are present, open data reproduce Clarivate's published self-citation shares within ten percentage points for 36 of 50 journals; the failures follow publishers that do not deposit references with Crossref, which also hides several stacking donors entirely. Open data contain a further artifact: biodiversity data downloads registered with DataCite inflate naive impact-factor numerators of taxonomy journals. The leading open method for stacking groups, CIDRE, detects none of 15 named Clarivate pairs. A diagnosis shows why: it flags the stacking citations but drops large donor journals, so a big journal feeding a small one never forms a group. A recipient-centric reading of CIDRE's own test, with its parameters unchanged, flags journals later listed for stacking 10 to 14 times more often than other journals, sometimes years before the listing; flagged journal pairs show more citations per citing paper and weaker topical similarity than matched pairs. Open data can support such audits, provided coverage gaps and their sources are reported alongside the results.

Keywords: citation stacking, journal self-citation, Journal Impact Factor, OpenCitations, research integrity, anomaly detection

1 Introduction

The Journal Impact Factor (JIF) counts the citations a journal receives in one year to the items it published in the two previous years. Because the JIF still shapes hiring, funding and journal choice, some journals inflate it, either by citing themselves or by receiving concentrated citations from a partner journal, a practice known as citation stacking (Heneberg 2016; Van Noorden 2013; Wilhite and Fong 2012; Chorus and Waltman 2016). Each year Clarivate withholds the JIF of journals whose citation patterns it judges anomalous and publishes the list, with key figures such as the share of self-citations in the JIF numerator or the share of a recipient's numerator that comes from one donor (Moussa 2022; Taşkın et al. 2021).

These decisions rest on Web of Science data, which outsiders cannot inspect. Open citation data now make independent checks possible in principle: OpenCitations releases citation links under open licences (Peroni and Shotton 2020; Heibi et al. 2019), and its Meta collection gives the bibliographic records behind them (Massari et al. 2024). Whether these data are good enough for integrity work is less clear. Coverage comparisons show that open sources approach the large commercial indexes on average (Martín-Martín et al. 2020; Visser et al. 2021; Culbert et al. 2025), but an average says little about the small, specific sets of journals that suppressions concern.

The best-known open method for this task is CIDRE (Kojaku et al. 2021). It removes citations that a degree-corrected stochastic block model can explain, scores journals as donors and recipients of the remaining "excessive" citations, and keeps groups of journals that pass a threshold. On Microsoft Academic Graph data it found 12 of 22 groups that Clarivate had suspended, and community detection found none.

We ask three questions. First, how faithfully do fully open data reproduce the figures Clarivate publishes, and where and why do they fail? Second, does CIDRE, run as published, detect the stacking cases Clarivate listed after its own validation period, and if not, why? Third, what do the excess citations look like at the level of individual papers? Our contributions are:

- a primary-source label set of Clarivate suppressions for data years 2015–2023 and an audit of open-data fidelity against Clarivate's published self-citation and stacking figures, with the failures traced to publishers' reference deposit and to a data-citation artifact;
- a diagnosis of a structural blind spot in CIDRE: its group-building step drops large donor journals, so the pattern Clarivate most often reports, a large journal feeding a small one, cannot form a group;
- a recipient-centric reading of CIDRE's own excessive-edge test that flags listed stacking journals well above chance, sometimes years before the listing, with a pre-specified evidence check of its flags;
- an analysis of the sensitivity of CIDRE to its stochastic null-model partition, and a paper-level anatomy of flagged journal pairs.

All analyses run on public data; code, run logs and checksums are released (Section 6).

2 Related work

Journal self-citation and citation stacking.

Coercive and excessive journal self-citation have been documented by surveys and large-scale analyses (Wilhite and Fong 2012; Chorus and Waltman 2016; Taşkın et al. 2021). Heneberg traced how excessive self-citation can shift into citation stacking between journals (Heneberg 2016), and a Brazilian citation scheme drew wide attention to stacking in 2013 (Van Noorden 2013). Repeated suppressions from the Journal Citation Reports have been analysed bibliometrically (Moussa 2022).

Detecting anomalous citation groups.

Fister et al. (2016) proposed searching citation networks for cartels, and unsupervised anomaly detection has been applied to journal-level networks (Jolly et al. 2020). CIDRE (Kojaku et al. 2021) is the reference method for journal groups validated against Clarivate's decisions. Paper-level approaches use graph learning (Liu et al. 2022), and recent work combines journal and paper levels in heterogeneous graphs (Liu et al. 2026). Barnes et al. (2026) show that journals flagged in the Journal Citation Reports receive citations from closer parts of the journal network.

Open data.

Coverage of OpenCitations and other open sources has been compared with commercial indexes (Martín-Martín et al. 2020; Visser et al. 2021; Culbert et al. 2025), but not, to our knowledge, in terms of what it means for detecting citation manipulation. Concurrent work by Zhang Qilin (2026), released as a dataset with code, compares 15 detection methods on OpenAlex for 1,057 journals in chemistry, materials science and nanoscience, using 16 Clarivate stacking pairs from 2018–2025. We differ in data source (OpenCitations), scope (all fields), the fidelity audit against Clarivate's published figures, the coverage diagnosis, and the analysis of CIDRE's mechanism.

3 Data and methods

3.1 Citation data and journals

We use the parsed OpenCitations snapshot of January 2024 released through the Illinois Data Bank (Mohasel Arjomandi et al. 2024), with 1,962,840,983 citation links between integer-identified publications. Journal (venue) assignments come from the OpenCitations Meta CSV dump (OpenCitations 2024; Massari et al. 2024), joined on the publication identifier; venue records are merged when they share an ISSN. Of 76.8 million journal articles, 72,720,703 obtain their venue from Meta, 388,394 from the venue field of the parsed file, and 3,695,762 have none. A venue counts as a journal if its own Meta record has type *journal* and it has an ISSN; this rule keeps 92,779 venues, of which 85,437 appear in the citation network. A stricter rule that also requires at least half of the venue's items to be journal articles is used as a sensitivity check. Processing used DuckDB (Raasveldt and Mühleisen 2019).

3.2 JIF-window counts and journal scores

For a citing year t, let $c(A \to B, t)$ be the number of citations from items of journal A published in t to items of journal B published in $t-1$ or $t-2$, the JIF window. We count citing items of any type but, following Clarivate's practice of counting citations from indexed sources, only citing items that have a journal; including citing items without a journal is a sensitivity analysis. For each recipient journal B and year t we compute the numerator $n_B(t) = \sum_A c(A \to B, t)$, the self-citation share $c(B \to B, t)/n_B(t)$, a JIF proxy dividing $n_B(t)$ by B's journal articles from $t-1$ and $t-2$, and the share of $n_B(t)$ from its largest external donor.

3.3 Labels: Clarivate suppressions 2015–2023

We rebuilt the suppression lists from primary Clarivate material: the "Suppressed Titles 2007–2017" spreadsheet (data years 2015–2017: journal, year and type), the Journal Citation Reports help pages for the 2019, 2022, 2023 and 2024 releases (archived copies), the 2020 release page and a copy of the 2021 release list. The resulting table has 144 rows: 98 self-citation suppressions, of which 60 come with a published self-citation share, 11 stacking journals without named partners (2015–2017), 17 named recipient–donor pairs with Clarivate's two stacking figures, one expression of concern with three donors, and six "suppression warnings" that we exclude. Journals were matched to our venues by ISSN taken from Crossref journal records, with a documented alias or ISSN choice where titles were ambiguous; 131 of 138 names matched.

3.4 Fidelity measures

Clarivate reports, for self-citation cases, the percentage of self-citations in the JIF numerator and, for stacking pairs, the donor's percentage of the recipient's JIF numerator and the "percentage exchange to JIF years", the share of all donor-to-recipient citations in year t that go to the JIF window [TODO: verify the verbatim JCR wording]. We compute the same quantities; for the exchange share our denominator counts cited years before t, so the estimate is an upper bound. We fixed in advance that a figure counts as reproduced if it lies within ± 10 percentage points, and that a listed journal "ranks" if it is in the top 1% of its year on the relevant share.

3.5 CIDRE

We implemented CIDRE as described by Kojaku et al. (2021), wrapping the authors' package and closing three gaps between package and paper: condition 1 (more than half of the $A \to B$ citations in year t go to the JIF window), the minimum of $\theta_w = 50$ within-group citations, and a compatibility fix. An edge is excessive if it meets condition 1 and its weight is significant under a Poisson test against the expected weight from a degree-corrected stochastic block model, with Benjamini–Hochberg correction at $\alpha = 0.01$ (Benjamini and Hochberg 1995). A journal's donor (recipient) score is the share of its outgoing (incoming) citations that are excessive within the current group;

journals whose donor and recipient scores are both below $\theta = 0.15$ are removed iteratively, and the remaining weakly connected components are the groups. The block model was fitted once on the network aggregated over all years with graph-tool's nonparametric inference (Peixoto 2017a,b). Detection follows the paper's rule: a Clarivate group is detected if at least half of a CIDRE group's journals and at least two journals are shared.

The block-model fit is stochastic. We therefore fitted seven partitions of the main journal network (five seeds, a repeat of the configuration seed, and a single-threaded fit) and one of the stricter journal rule, and ran CIDRE on every year from 2015 to 2023 with each.

3.6 A recipient-centric reading of CIDRE

Clarivate's stacking criterion is recipient-centric: a recipient that depends on one donor for a large share of its numerator. CIDRE's excessive-edge test can be read the same way, without the group step. For journal j in year t, let e_j be the number of excessive citations it receives (self-citations excluded) and s_j its in-strength. We flag j if $e_j/s_j \geq \theta$ and $e_j \geq \theta_w$, with CIDRE's own $\theta = 0.15$ and $\theta_w = 50$. This variant was designed after we diagnosed CIDRE's misses on two 2023 cases (Section 4.3); those two journals are therefore excluded from every evaluation in which they could favour it. Because flags vary between partitions, we also report consensus flags present in at least k of the seven partitions. We evaluate the variant by enrichment: the share of listed journals ever flagged in 2015–2023 compared with the share of all journals with at least 20 numerator citations in some year, with a one-sided binomial test.

3.7 Paper-level anatomy and evidence check

For flagged recipients and their largest excessive donor, and for named Clarivate pairs, we collect the citing papers behind the pair's window citations and compute the citations per citing paper, the Gini coefficient of citations over citing papers, and the shares coming from the single top citing paper and from the top 10% of citing papers. We compare titles of citing and cited papers by TF–IDF cosine similarity. Each pair receives up to three controls from the same year with 0.8 to 1.25 times its window citations, both journals in the journal rule, and a recipient never flagged. As a structured evidence check, fixed before the sample was drawn, we record three public indicators per pair: either journal appears in Clarivate's lists; recipient and donor share a Crossref publisher; or the top 10% of citing papers carry at least half of the pair's citations.

3.8 Reproducibility

Every analysis is a script that writes a run folder with its parameters, input and output checksums and log, and the decisions and pass criteria were recorded in a lab notebook before the corresponding runs. We distinguish pre-specified from post hoc analyses throughout.

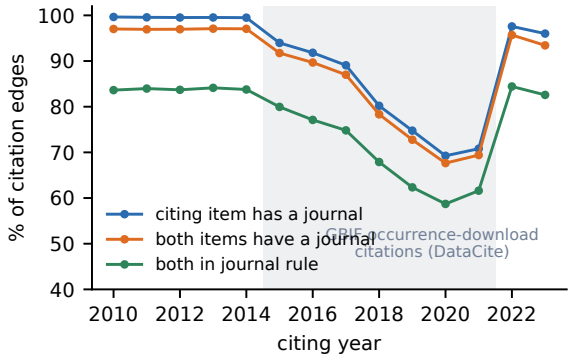


Fig. 1 Share of citation links whose citing item has a journal, whose both items have a journal, and whose both items are journals under the journal rule, by citing year. The 2015–2021 dip is due to GBIF occurrence-download records registered with DataCite.

3.9 Use of generative AI

[TODO: author to review and confirm wording and model versions] Claude (Anthropic) was used as a coding and writing assistant: to write and run analysis scripts under the author's direction, to search the literature, and to draft and edit the text. All analyses, numbers and references were checked by the author against the run records and the original sources (DOIs), and the author takes full responsibility for the content.

4 Results

4.1 What open data contain

In citing years up to 2014, 99.5% of citation links have a citing item with a journal; this share falls to 69.3% in 2020 and recovers to 97.6% in 2022 (Fig. 1). The dip is caused by one source. In 2020, 221,918 of the 230,438 citing items typed *dataset* without a journal (96.3%) carry DOIs of the Global Biodiversity Information Facility (prefix 10.15468, registered with DataCite): occurrence downloads whose reference lists cite the datasets and taxonomic publications the records come from. These are legitimate data citations, but they are not the journal citations Clarivate counts. Had they been counted, they would form 12.8%, 14.4% and 11.2% of the JIF-window numerator in 2019, 2020 and 2021, concentrated in a few taxonomy journals: in 2020, 99.8% of the numerator of *Zootaxa* would come from such items, giving a JIF proxy above 600. Restricting citing items to those with a journal removes the artifact; including them changes the JIF proxy noticeably only in 2019–2021 (correlation with the main version 0.70–0.90, against at least 0.99 in other years).

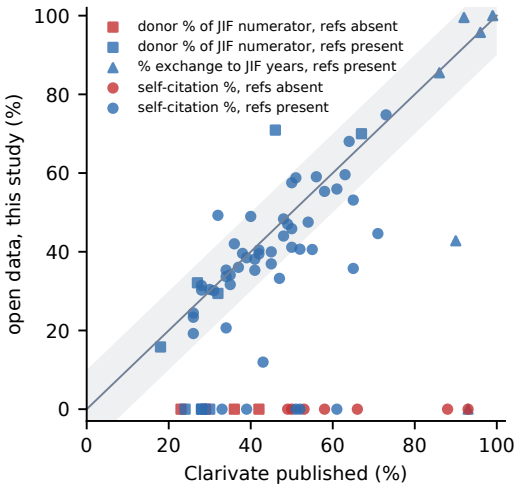


Fig. 2 Clarivate's published percentages against ours, by whether the citing journal's (self-citation) or donor's (stacking) reference lists are present in OpenCitations. The band marks ±10 percentage points.

4.2 Fidelity to Clarivate's published figures

Whether open data reproduce Clarivate's figures depends mainly on whether the citing journal's reference lists are in OpenCitations (Fig. 2). When they are, our self-citation share lies within ±10 percentage points of Clarivate's for 36 of 50 journals, with a median absolute difference of 4.6 points and a median signed difference of −2.8 points. When they are absent, our share is zero and none of seven journals passes. For stacking pairs whose donor has references and the pair has citations in our data (five pairs), the donor's share of the recipient's numerator is reproduced for four and the exchange share for four. Many stacking donors in Clarivate's lists, however, have no references in OpenCitations at all.

The missing references are not random. For the journals whose references are absent, the median share of their DOIs with deposited reference lists in Crossref is zero (zero for eight of eleven), against 0.96 for journals whose references are present. The journals of one publisher that deposits references today began only recently (backfile coverage around 5%). These blind spots include donors in several Clarivate stacking cases.

On ranks, 59 of 85 self-citation suppressions (69%) lie in the top 5% of their year by self-citation share, with a median percentile of 3.1%, but only 9 are in the top 1%. The criterion we fixed in advance for data year 2020, at least half of the suppressed journals in the top 1%, was not met (0 of 8); we report this as recorded.

4.3 CIDRE on open data

CIDRE finds between 11 and 32 groups per year, typically of three to five journals. It detects none of the 15 named Clarivate pairs from 2018–2023 for which both journals are matched, in any of the eight partitions, and no recipient or donor of these pairs

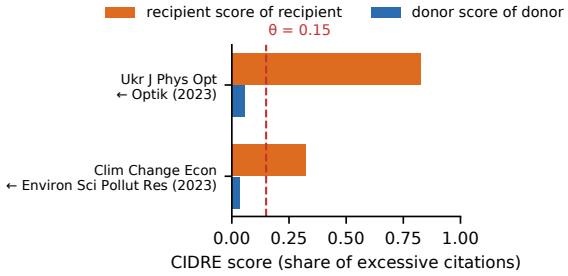


Fig. 3 Why CIDRE misses two well-covered 2023 stacking pairs: the recipients' scores exceed θ, but the large donors' scores do not, so the group-building step removes the donors.

is in any CIDRE group in its year. Some misses are due to data: the donor has no references in OpenCitations, or the pair has too few citations to reach θ_w. Two 2023 pairs, however, are well covered and still missed.

For these, a stage-by-stage diagnosis shows the mechanism (Fig. 3). The stacking edge passes CIDRE's test: 99.5% (Optik to Ukrainian Journal of Physical Optics) and 85.5% (Environmental Science and Pollution Research to Climate Change Economics) of the donor's citations to the recipient fall in the JIF window, and the edges are significant against the block model. The recipients' scores, 0.83 and 0.32, are well above θ. But the donors are large journals whose stacking citations are a small part of their output: their donor scores are 0.057 and 0.035. Since CIDRE removes a journal when both of its scores are below θ, the donor is removed, the recipient's score falls to zero, and no group remains. The CIDRE paper motivates $\theta = 0.15$ on the recipient side, noting that all recipients suspended by Clarivate received at least 15% of their citations from donors (Kojaku et al. 2021), but applies the same threshold to donors.

CIDRE does find stacking between small journals. In 2015 one of its groups consists exactly of Applied Clinical Informatics and Methods of Information in Medicine, and in 2017 one consists exactly of Liver Cancer and Digestive Diseases; in each case both journals were suppressed for stacking in that year, although Clarivate's spreadsheet for those years does not name the pairs.

4.4 The recipient-centric reading

Over 2015–2023, the variant flags 12 of 30 stacking-listed journals (40%), against 3.4% of all journals with at least 20 numerator citations in some year: an enrichment of 11.7 ($p = 1.2 \times 10^{-10}$). Without the two journals used to design it, 10 of 28 are flagged (10.5-fold, $p = 1.6 \times 10^{-8}$), and across partitions and size thresholds the enrichment ranges from 5.7 to 10.5 ($p \leq 1.1 \times 10^{-5}$). For self-citation-listed journals the enrichment is weaker and not robust across partitions (1.3 to 3.4), as expected for a measure of external citations.

Requiring flags to appear in more partitions lowers the base rate faster than the hit rate (Fig. 4). With flags present in all seven partitions, 2.3% of journals are ever flagged, against 9 of 28 stacking-listed journals: a 13.9-fold enrichment ($p = 8.7 \times 10^{-9}$). Because the consensus rule was chosen after seeing these results, we report all three levels.



Fig. 4 Share of listed journals ever flagged by the recipient-centric reading in 2015–2023, against the base rate, by the number of partitions required (design cases excluded). Labels give the enrichment for stacking-listed journals.

Eight listed journals were flagged before their first listing (Fig. 5). Granular Computing was flagged in every year from 2016 to 2020, three to seven years before its 2023 listing, with Information Sciences, its donor in Clarivate's list, as the largest excessive donor in four of those years. Panminerva Medica and Minerva Medica were flagged three to four years before their 2023 listing, Liver Cancer one to two years before 2017 (largest excessive donor Digestive Diseases), and Ukrainian Journal of Physical Optics in 2022 (donor Optik).

In the pre-specified evidence check on 196 randomly sampled consensus flags and 196 matched controls, 28.1% of flagged pairs carry at least one independent indicator against 6.1% of controls ($p = 6.2 \times 10^{-9}$, Fisher's exact test). The most frequent indicator is a shared publisher of recipient and donor (19.4% against 6.1%); 5.1% of flagged pairs have a donor in Clarivate's lists and 5.6% have strongly concentrated citations, against none of the controls. These shares are a lower bound on the precision of the flags, not an estimate of it.

4.5 Paper-level anatomy

Flagged pairs differ from matched controls in how their citations arise (Fig. 6). Each citing paper cites the recipient 1.98 times on median, against 1.28 ($p = 2.5 \times 10^{-27}$, Mann–Whitney), and citations are more unevenly spread over citing papers (Gini 0.25 against 0.18, $p = 1.5 \times 10^{-8}$). The share from the top 10% of citing papers does not differ ($p = 0.35$): the pattern is many papers each citing a little too often rather than a few papers with long reference lists. Cited and citing titles are less similar in flagged pairs (TF–IDF cosine 0.084 against 0.133, $p = 1.2 \times 10^{-24}$). Since excessive edges are defined against the block model, flagged pairs might simply join more distant fields; in fact 27.9% of flagged pairs share a block against 0.6% of controls, and within cross-block pairs the differences remain (cosine 0.084 against 0.133, $p = 1.2 \times 10^{-17}$; citations per citing paper 2.00 against 1.28). For the five named Clarivate pairs with citations the sample is too small for conclusions.

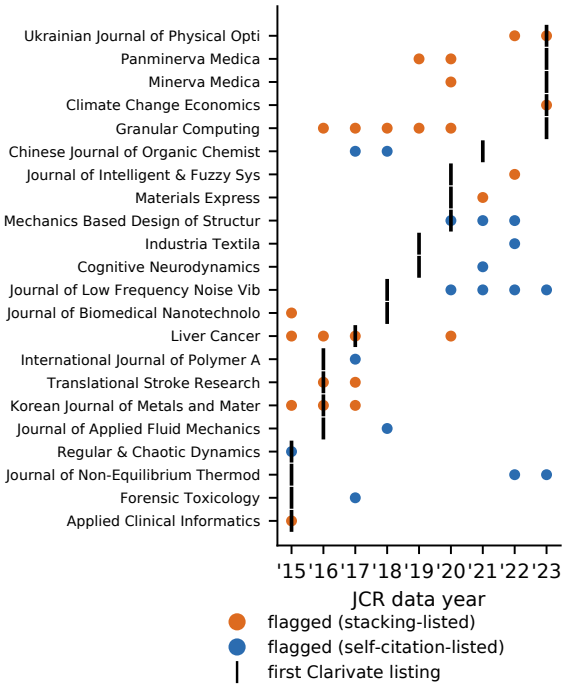


Fig. 5 Years in which listed journals were flagged (dots) and their first Clarivate listing (bars), partition *main*.

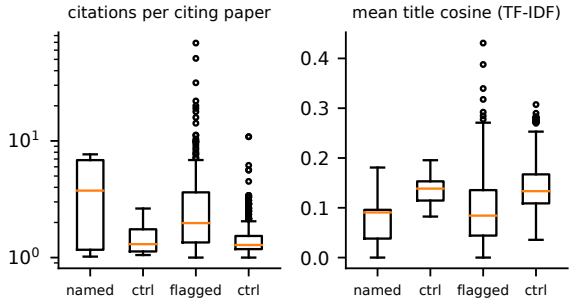


Fig. 6 Citations per citing paper (log scale) and citing–cited title similarity for named Clarivate pairs, flagged pairs and their matched controls.

4.6 Sensitivity to the block-model partition

With eight threads, block-model fits are not reproducible from a seed: six fits of the same network give 2,131 to 2,912 blocks, with pairwise normalized mutual information of 0.76–0.78 and adjusted Rand index (Hubert and Arabie 1985) of 0.44–0.54. Two single-threaded fits with the same seed give identical partitions, so the variation comes from the parallel search. CIDRE's groups depend much less on the partition than the

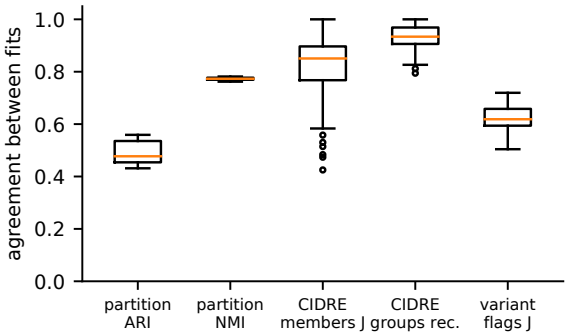


Fig. 7 Agreement between fits: block-model partitions (ARI, NMI), CIDRE group members (Jaccard) and group recovery, and the variant's flags (Jaccard).

partitions themselves agree: the median Jaccard similarity of group members between partitions is 0.85, and 93% of groups have a counterpart (Fig. 7). The variant's flags are less stable (median Jaccard 0.62), which motivates the consensus rule.

5 Discussion

What open data can and cannot show.

Open citation data reproduce Clarivate's self-citation figures closely when reference lists are present, so open audits of journal citation behaviour are feasible. But the gaps sit where integrity questions arise: some publishers deposit no references, and several of the donors behind Clarivate's stacking cases are among them. An open audit can therefore not conclude that a journal shows no anomaly when its citing partners are absent from the data; coverage of the relevant journals should be reported alongside any result. Data citations from repositories, such as GBIF downloads, are a second, newer source of distortion for anyone computing impact-factor-like indicators from open data.

Why CIDRE misses Clarivate-style stacking.

CIDRE was designed to find groups whose members exchange excessive citations, and it does find stacking between small journals. Its symmetric threshold, however, requires a donor to devote a substantial share of its own citations to the group. In the stacking cases Clarivate reported in 2023, large journals fed small ones, and CIDRE flags the edge but discards the group. This is a property of the method, not of the data: it appears with well-covered donors. A recipient-centric reading of the same test recovers listed stacking journals far above chance and often before listing, at the cost of flagging about 2–5% of journals over nine years, depending on the consensus level.

What the flags mean.

The flags mark anomalous citation patterns, not misconduct. Many have benign explanations, such as a society journal and its companion, editorial preferences, or regional

communities. The evidence check shows that flagged pairs carry independent indicators far more often than matched pairs, and the paper-level anatomy shows that their citations are spread over many papers that cite the recipient repeatedly and are topically further from it. Flags are candidates for expert review, not verdicts.

Limitations.

Clarivate names few stacking pairs, so recall figures rest on a small number of cases, and our enrichment results rely on 28 to 30 stacking-listed journals. The recipient-centric reading and its consensus rule were chosen after part of the evaluation; we excluded the cases used in the design and report all variants, but a test on future suppression lists is needed. The evidence check is a lower bound on precision. Clarivate's exact formulas and thresholds are not public, and our exchange share is an upper bound. Crossref coverage statistics describe the present, not the state at the time of the snapshot. Title similarity uses TF–IDF on titles only, and controls are matched on year and volume, not on field.

6 Conclusions

Fully open citation data can reproduce much of what Clarivate publishes about its journal suppressions, and they can anticipate some of its stacking decisions. Two conditions matter: auditors must check that the relevant reference lists are present, and methods must match the pattern they target. A large journal feeding a small one is invisible to CIDRE's group definition but visible to a recipient-centric reading of the same statistical test. We release the labels, code and run records so that the audit can be repeated on new suppression lists.

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- **Competing interests:** [TODO: e.g. "The authors have no competing interests to declare."]
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** Citation data: Mohasel Arjomandi et al. (2024); venue metadata: OpenCitations (2024). Suppression labels with their archived Clarivate sources, all scripts, configuration, run logs and checksums: [TODO: repository URL and DOI].
- **Code availability:** See Data availability.
- **Author contributions:** [TODO: authors]

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